

Full Assessment Question Paper –Web Development (HTML, CSS, JavaScript & React)

Duration: 3 Hours

Total Marks: 100

Section A – HTML & CSS (20 Marks)

Part A – Multiple Choice Questions (1 Mark Each)

1. Which tag is used to create a hyperlink?
 - a) <link>
 - b) <a>
 - c) <href>
 - d) <url>
2. Which HTML tag is used for creating a form?
 - a) <input>
 - b) <form>
 - c) <table>
 - d) <button>
3. Which CSS property changes text color?
 - a) background-color
 - b) text-color
 - c) color
 - d) font-color
4. Which CSS layout system is one-dimensional?
 - a) Grid
 - b) Flexbox
 - c) Table
 - d) Float
5. Which media query is used for mobile devices?
 - a) @media screen and (max-width:768px)

- b) @media print
 - c) @media all
 - d) None
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Part B – Short Answer Questions (2 Marks Each)

6. Explain semantic HTML tags with examples.
 7. Differentiate between inline, internal, and external CSS.
 8. What is the CSS Box Model?
 9. Explain Flexbox and its advantages.
 10. What is Responsive Web Design?
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Part C – Practical Task (10 Marks)

Create a Student Registration Form with:

- Name
- Email
- Mobile Number
- Gender
- Course Selection
- Submit Button

Apply:

- External CSS
 - Responsive Layout
 - Hover Effects
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Section B – JavaScript (30 Marks)

Part A – Multiple Choice Questions (1 Mark Each)

11. Which keyword declares a block-scoped variable?
 - a) var

- b) let
- c) static
- d) public

12. Which method adds an element to an array?

- a) push()
- b) pop()
- c) shift()
- d) splice()

13. Which function is used to fetch API data?

- a) get()
- b) request()
- c) fetch()
- d) load()

14. Which loop executes at least once?

- a) for
- b) while
- c) do-while
- d) foreach

15. Which operator checks both value and type?

- a) ==
- b) =
- c) ===
- d) !=

Part B – Coding Questions (3 Marks Each)

16. Write a program to find the largest number in an array.

17. Write a function to reverse a string.

18. Create a Grade Calculator using if-else.

19. Write a program using a loop to print numbers from 1–100.
 20. Create a Student Object with name, age, and course.
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Part C – DOM & Events (10 Marks)

21. Create a webpage where clicking a button changes text content.
 22. Build a form validation page for:
 - Name
 - Email
 - Password
 23. Create a Dynamic List:
 - Add Item
 - Remove Item
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Section C – React JS (30 Marks)

Part A – Theory Questions (2 Marks Each)

24. What is React?
 25. What is JSX?
 26. What are Components?
 27. Difference between Props and State.
 28. Explain useEffect Hook.
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Part B – Coding Questions (4 Marks Each)

29. Create a Counter Application using useState.
 30. Create a Login Form using React.
 31. Display Student Data using Props.
 32. Create a Dark Mode Toggle using Context API.
 33. Fetch User Data from API using useEffect.
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Part C – React Router (5 Marks)

34. Create a React Application with:

- Home Page
- About Page
- Contact Page

Use React Router for navigation.

Section D – Project & Real-Time Scenario (20 Marks)

Project Design Questions

35. Design an E-Commerce Frontend Application:

- Product Listing
- Search
- Filter
- Add to Cart

Explain folder structure and components.

(10 Marks)

36. Student Management System

Develop a React Application with:

- Add Student
- Edit Student
- Delete Student
- View Student List

Features:

- Routing
- Form Validation
- State Management

(10 Marks)

Viva / Interview Questions

HTML/CSS

1. Difference between ID and Class.
2. What are semantic tags?
3. Explain Flexbox and Grid.
4. What is z-index?
5. Difference between margin and padding.

JavaScript

6. Difference between var, let, and const.
7. What is hoisting?
8. Difference between == and ===.
9. What is closure?
10. What is callback function?

React

11. What is Virtual DOM?
12. What are Hooks?
13. What is useState?
14. What is Context API?
15. Why use React Router?

Final Capstone Assessment (30 Marks – Practical)

Choose Any One Project:

Option 1: Job Portal

- Job Listing
- Search Jobs
- Apply Button
- Responsive Design

Option 2: Food Delivery UI

- Restaurant List
- Food Menu
- Cart Page
- Checkout UI

Option 3: Bus Tracking UI

- Bus List
- Route Details
- Search Bus
- Responsive Dashboard

Evaluation Criteria

Criteria	Marks
UI Design	5
HTML/CSS	5
JavaScript Logic	5
React Components	5
Routing	3
API Integration	3
Code Quality	2
Responsive Design	2
Total	30

Grand Total: 100 Marks